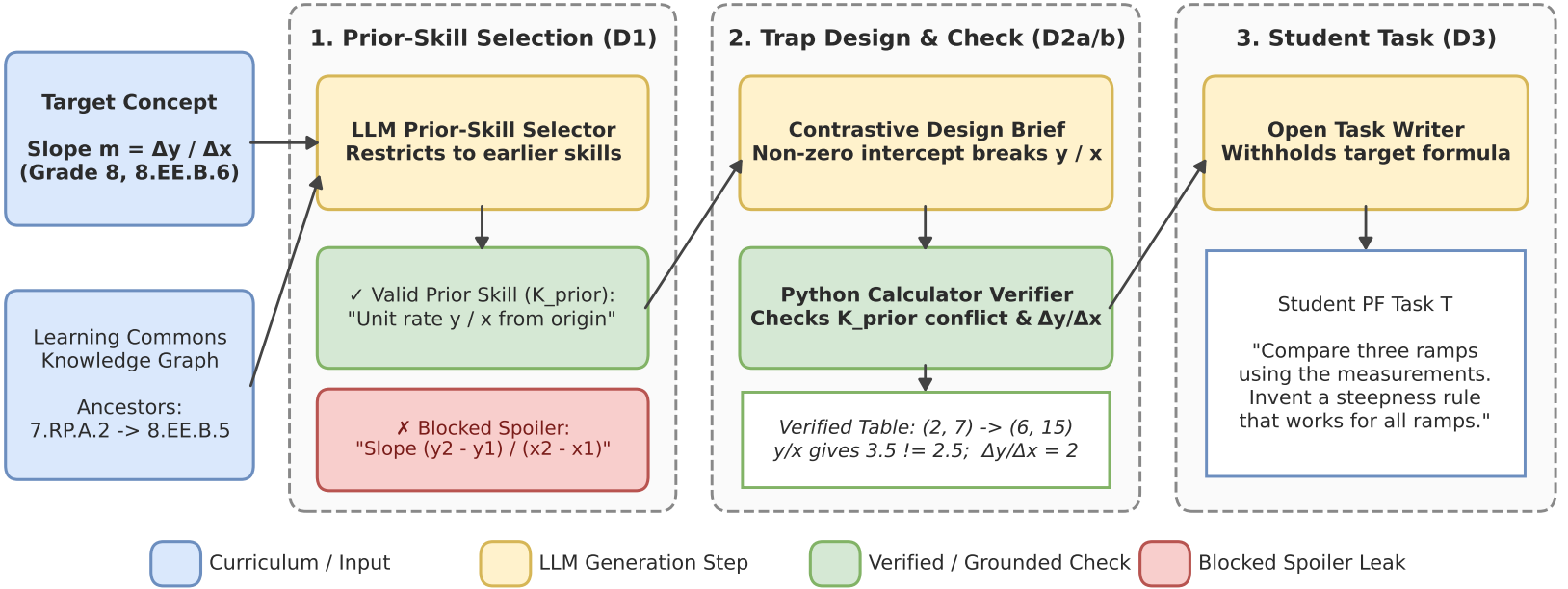
Target Concept
Slope m = Δy / Δx
(Grade 8, 8.EE.B.6)
Learning Commons Knowledge Graph
Ancestors:
7.RP.A.2 -> 8.EE.B.5
1. Prior-Skill Selection (D1)
LLM Prior-Skill Selector
Restricts to earlier skills
✓ Valid Prior Skill (K_prior):
"Unit rate y / x from origin"
✗ Blocked Spoiler:
"Slope (y2 - y1) / (x2 - x1)"
2. Trap Design & Check (D2a/b)
Contrastive Design Brief
Non-zero intercept breaks y / x
Python Calculator Verifier
Checks K_prior conflict & Δy/Δx
Verified Table: (2, 7) -> (6, 15)
y/x gives 3.5 != 2.5; Δy/Δx = 2
3. Student Task (D3)
Open Task Writer
Withholds target formula
Student PF Task T
"Compare three ramps using the measurements. Invent a steepness rule that works for all ramps."
Curriculum / Input
LLM Generation Step
Verified / Grounded Check
Blocked Spoiler Leak